

① Taller:

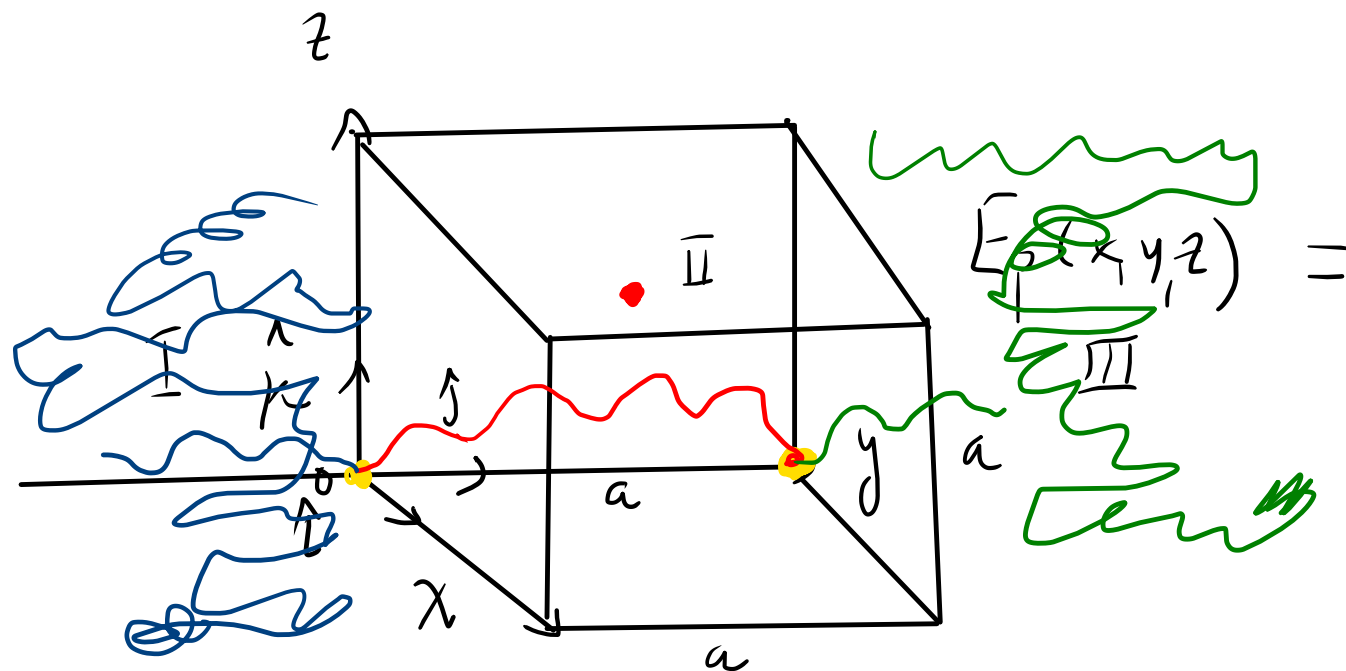
• Electrón Confinado en un potencial 3D

* Autofunciones ✓

* Autovalores ✓

* Degeneración ✓

* $E_f - E_i = h\nu$ ✓



$$\left\{ \begin{array}{l} 0 \leq x \leq a \\ 0 \leq y \leq a \\ 0 \leq z \leq a \\ \infty \quad x < 0, y < 0, z < 0 \\ \quad \quad x > a, y > a, z > a \end{array} \right.$$

② • Resolver ecuación de Schrödinger

$$\hat{H}\phi = E\phi; \quad \{\phi_1, \phi_2, \dots, \phi_n\}$$

$$\{E_1, E_2, \dots, E_n\}$$

$$\hat{H} = \frac{\hat{P}}{2me} = -\frac{\hbar^2 \nabla^2}{2me}$$

$$\hat{H}\phi_{\pi} = E\phi_{\pi}; \quad -\frac{\hbar^2 \nabla^2 \phi_{\pi}}{2me} = E\phi_{\pi}$$

$$\textcircled{3} \quad \nabla^2 \phi_{II} + K^2 \phi_{II} = 0, \quad K^2 = \frac{2me}{\hbar^2} E$$

Separación de Variable

$$\phi_{II}(x, y, z) = f(x) g(y) h(z)$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (f(x) g(y) h(z)) + K^2 f g h = 0$$

$$\underbrace{g(y) h(z)}_{\partial x^2} \frac{\partial^2 f}{\partial x^2} + \underbrace{f(x) h(z)}_{\partial y^2} \frac{\partial^2 g}{\partial y^2} + \underbrace{f(x) g(y)}_{\partial z^2} \frac{\partial^2 h}{\partial z^2} + K^2 f g h = 0$$

multiplico por $\frac{1}{\phi} = \frac{1}{f g h}$

$$\frac{1}{f} \frac{d^2 f}{dx^2} + \frac{1}{g} \frac{d^2 g}{dy^2} + \frac{1}{h} \frac{d^2 h}{dz^2} + K^2 = 0$$

$$-K_x^2$$

$$-K_y^2$$

$$-K_z^2$$

$$K^2 = K_x^2 + K_y^2 + K_z^2$$

$$\textcircled{1} \quad \frac{d^2 f}{dx^2} + K_x^2 f = 0 \quad \textcircled{1}$$

$$\frac{d^2 g}{dy^2} + K_y^2 g = 0 \quad \textcircled{2}$$

$$\frac{d^2 h}{dz^2} + K_z^2 h = 0 \quad \textcircled{3}$$

$$f(x) = A \sin(K_x x) + B \cos(K_x x)$$

$$g(y) = C \sin(K_y y) + D \cos(K_y y)$$

$$h(z) = E \sin(K_z z) + F \cos(K_z z)$$

$$\phi_{\text{II}}(x, y, z) = (A \sin(K_x x) + B \cos(K_x x)) (C \sin(K_y y) + D \cos(K_y y)) (E \sin(K_z z) + F \cos(K_z z))$$

$$\phi_{\text{II}}(0, y, z) = 0$$

$$= B g(y) h(z) = 0$$

$$B = 0 \quad \checkmark$$

$$\phi_{\text{II}}(x, y, z) = A \sin(K_x x) g(y) h(z)$$

$$\phi_{\text{II}}(x, y, z) = 0$$

$$A \sin(K_x a) g(y) h(z) = 0$$

$$\sin(K_x a) = 0; \quad K_x a = n_x \pi$$

$$K_x = \frac{n_x \pi}{a} \quad n_x: \text{entero}$$

$$\phi_{\text{II}}(x, y, z) = A \sin\left(\frac{n_x \pi}{a} x\right) g h$$

$$\phi_{\text{II}}(x, y=0, z) = 0$$

$$\phi_{\text{II}}(x, y=a, z) = 0$$

$$\phi_{\text{II}}(x, y, 0) = 0$$

$$\phi_{\text{II}}(x, y, a) = 0$$

$$\phi_{\text{II}}(x, y, z) = A \sin\left(\frac{n_x \pi}{a} x\right) C \sin\left(\frac{n_y \pi}{a} y\right) E \sin\left(\frac{n_z \pi}{a} z\right)$$

$$\phi_{\text{II}}(x, y, z) = M \sin\left(\frac{n_x \pi}{a} x\right) \sin\left(\frac{n_y \pi}{a} y\right) \sin\left(\frac{n_z \pi}{a} z\right)$$

$$M = A C E; \quad n_x, n_y, n_z: \text{enteros}$$

Condición de normalización:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |\phi|^2 dx dy dz = 1$$

$$\int_0^a \int_0^a \int_0^a |\phi|^2 \sin^2\left(\frac{n_x \pi}{a} x\right) \sin^2\left(\frac{n_y \pi}{a} y\right) \sin^2\left(\frac{n_z \pi}{a} z\right) dx dy dz = 1$$

$$\left(\int_0^a \left(\int_0^a \sin^2\left(\frac{n_x \pi}{a} x\right) dx \right) \sin^2\left(\frac{n_y \pi}{a} y\right) \sin^2\left(\frac{n_z \pi}{a} z\right) dy dz \right) = \frac{1}{|M|^2}$$

$$\int_0^a \sin^2\left(\frac{n_x \pi}{a} x\right) dx = \int_0^a \left(\frac{1}{2} - \frac{\cos\left(\frac{2n_x \pi}{a} x\right)}{2} \right) dx$$

$$= \int_0^a \frac{1}{2} dx - \int_0^a \frac{1}{2} \cos\left(\frac{2n_x \pi}{a} x\right) dx$$

$$\textcircled{5} \int_0^a \sin^2\left(\frac{n_x \pi}{a} x\right) dx = \frac{a}{2}$$

$$\frac{1}{|M|^2} = \frac{a}{2} \frac{a}{2} \frac{a}{2} = \frac{a^3}{2^3}$$

$$M = \left(\frac{2}{a}\right)^{3/2} \checkmark$$

$$\phi(x, y, z) = \left(\frac{2}{a}\right)^{3/2} \sin\left(\frac{n_x \pi}{a} x\right) \sin\left(\frac{n_y \pi}{a} y\right) \sin\left(\frac{n_z \pi}{a} z\right)$$

$$\phi_{n_x n_y n_z}(x, y, z) = \left(\frac{2}{a}\right)^{3/2} \sin\left(\frac{n_x \pi}{a} x\right) \sin\left(\frac{n_y \pi}{a} y\right) \sin\left(\frac{n_z \pi}{a} z\right)$$

$$\{ \phi_{111}, \phi_{121}, \phi_{211}, \phi_{112}, \dots \}$$

⑤ Autovalores de Energía

$$K^2 = K_x^2 + K_y^2 + K_z^2$$

$$K^2 = \frac{2m_e E}{\hbar^2}$$

$$K_x = \frac{n_x \pi}{a}, \quad K_y = \frac{n_y \pi}{a}, \quad K_z = \frac{n_z \pi}{a}$$

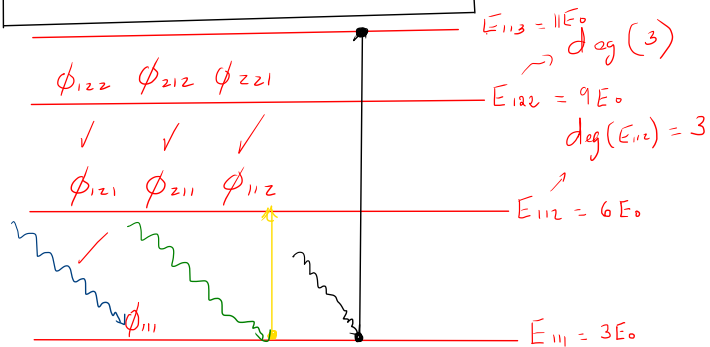
$$\frac{2m_e E}{\hbar^2} = \frac{n_x^2 \pi^2}{a^2} + \frac{n_y^2 \pi^2}{a^2} + \frac{n_z^2 \pi^2}{a^2}$$

$$E = \frac{\hbar^2}{2m_e} \frac{\pi^2}{a^2} (n_x^2 + n_y^2 + n_z^2)$$

$$E_{n_x n_y n_z} = \frac{\hbar^2 \pi^2}{2m_e a^2} (n_x^2 + n_y^2 + n_z^2)$$

$$E_0 = \frac{\hbar^2 \pi^2}{2m_e a^2}$$

$$E_{n_x n_y n_z} = (n_x^2 + n_y^2 + n_z^2) E_0$$



$$E_f - E_i = h\nu$$

$$6E_0 - 3E_0 = h\nu; \quad 3E_0 = h\nu$$

$$E_0 = \frac{\hbar^2 \pi^2}{2m_e a^2}$$

a : Dimension

$$\frac{3\hbar^2 \pi^2}{2m_e a^2} = h\nu$$

$$a = 1 \text{ nm} = 10^{-9} \text{ m}$$

$$a = 1 \text{ Å} = 10^{-10} \text{ m}$$

$$a = 1 \mu\text{m} = 10^{-6} \text{ m} \quad \checkmark$$

$$a = 1 \text{ mm} = 10^{-3} \text{ m}$$

$$a = 1 \text{ m}$$

$$m_e = 0.511 \frac{\text{MeV}}{c^2}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$\deg(E_{123}) = 6$$

$$\underline{\phi_{123} \quad \phi_{132} \quad \phi_{231} \quad \phi_{213} \quad \phi_{312} \quad \phi_{321}} \quad E_{123} = 14 \bar{E}_0$$

$$\underline{\phi_{222}} \quad E_{222} = 12 \bar{E}_0$$